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REVIEW OF THE PAPER :

Volatility is rough

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1 Introduction

Most famous and common models for volatility are based on dynamics driven by Brownian motion. The use of Brownian motion for the dynamic leads to volatility process with smoothness close to the Brownian motion one. This high level of smoothness and regularity isn't compatible with some accepted stylized facts (e.g long memory, ATM decay). In order to obtain volatility process with different level of smoothness a new class of model based on dynamics involving fractional Brownian motion got introduced in the late 90's. This paper introduces a model driven by fractional Brownian motion with a smoothness paramater close to 0.

2 Fractional Brownian Motion

2.1 Definition

A fractional Brownian motion(fBm) with parameter $H(H \in]0,1[)$ is a real centered Gaussian process noted B_t^H, $t \in \mathbb{R}$, defined on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$, such that :

— $B_0^H = 0$ a.s

— Its covariance function is defined by :

$$pH : \mathbb{R}^2 \mapsto \mathbb{R}$$

$$(t,s) \mapsto \frac{1}{2}(|s|^{2H} + |t|^{2H} - |t-s|^{2H})$$

where H is called the **Hurst index**.

Remark : In contrary to the regular Brownian motion, the increments of the fractional Brownian motion are determined by the Hurst index :

— if $H = \frac{1}{2}$, the increments are independ and the process is a regular Brownian motion ;

— if $H > \frac{1}{2}$, the increments are positively correlated ;

— if $H < \frac{1}{2}$, the increments are negatively correlated.

2.2 Propreties

1. **Self-similarity** : The fBm movement have the same behavior in each time interval if we choose the spatial scale correctly : $\forall a > 0$ and $\forall t \in \mathbb{R}, B_{at}^H \overset{\mathcal{L}}{=} a^H B_t^H$.
   Proof :
   $$\forall\ (t,s) \in \mathbb{R}^2,\ \mathbb{E}[B_{at}^H B_{as}^H] = \frac{1}{2}(|as|^{2H} + |at|^{2H} - |at-as|^{2H})$$
   $$= \frac{1}{2}a^{2H}(|s|^{2H} + |t|^{2H} - |t-s|^{2H})$$
   $$= a^{2H}\mathbb{E}[B_t^H B_s^H]$$
   $$= \mathbb{E}[a^H B_t^H a^H B_s^H]$$
2. **Stationary increments** : $\forall h > 0, (B_{t+h}^H - B_h^H)_{t\geq 0} \overset{\mathcal{L}}{=} (B_t^H)_{t\geq 0}$
3. **Gaussian process** : with $\mathbb{E}[W_1^H] = 0$ and $\mathbb{E}[(W_1^H)^2] = 1$
4. **Not markovian** : Since, the fBm is not Markovian, it is difficult to study and to put these models into practice.
   Moreover, $H \neq \frac{1}{2} \Rightarrow, W^H$ is not a semimartingale. So we can not apply ItÃ´'s calculus. (We can apply pathwise techniques (ZÃ¤hle (1998)) and Malliavin calculus techniques (Carmona, Countin and Montseny (2003)).
5. **Continuity** : We have seen that the fBm has continous modification (using Kolmogorov continuity theorem, see Appendix). So we assume that :$\forall \varepsilon > 0, B^H$ is $(H-\varepsilon)$-Holder a-s.

Reminder : a function with a Holder constante less than one, is continuous but not derivable.

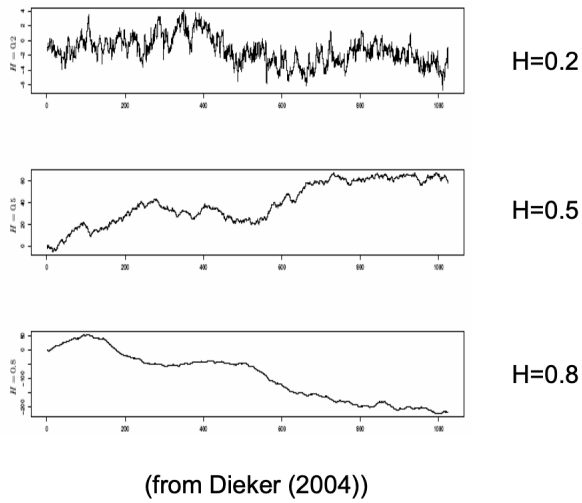

FIGURE 1 – Paths of fBm for different values of H

2.3 Representation

Mandelbrot-Van Ness :
According to **Mandelbrot** and **Van Ness** (1968), fractional Brownian motion admits the following representation :

$$\forall t, B^H(t) = B^H(0) + \frac{1}{\Gamma(H+\frac{1}{2})}\left(\int_{-\infty}^{0}[(t-s)^{H-\frac{1}{2}} - (-s)^{H-\frac{1}{2}}]dB(s) + \int_0^t (t-s)^{H-\frac{1}{2}}dB(s)\right) \tag{1}$$

where B is a standard Brownion motion.

Riemann-Louville representation :
Another way to highlight the fBm is the following representation :

$$W_t^H = \int_0^t K_H(t-s)dW_s, \quad K_H(t) = \frac{t^{(H-1/2)}}{\Gamma(H+1/2)}, \quad H \in (0,1)$$

2.4 Utility in Finance

Motivated by empirical studies, several authors have studied fnancial models driven by the fBm.
For instance, **Fractional stochastic volatility models** (Comte and Renauld (1998, 2003), explain better the long-time behaviour of the implied volatility.

Another example, is the **Fractional Black-Scholes model**, where the Bm in the BS model is replaced by a fBm (Mandelbrot (1997), Willinger (1999)). Unfortunately, they allow for arbitrage opportunities (Sottinen (2001)). But this arbitrage opportunities disappear by introducing minimal amount of time between transactions, or under transaction costs (see Guasoni (2005)).

3 Smoothness of the volatility

In the first part of this section we assume that we are able to observe directly the volatily (or log volatily) process. This is a strong assumption that is of course not true in reality. We are going to use estimated value

for the volatility when doing our numerical computation. This assumption will be discussed in the second part of this section.

3.1 The estimator

We try to estimates the smsoothness of the volatility process, by taking a look at the momentum function of our obversations of the volatility process on a grid mesh Δ on $[0, T]$:

$$m(q, \Delta) = \frac{1}{N} \sum_{k=1}^{N} |\log(\sigma_{k\Delta}) - \log(\sigma_{(k-1)\Delta})|^q \tag{2}$$

Assuming the increments of the log-volatility process are stationary and that a law of large number can be applied, $m(q, \Delta)$ can also be seen as the empirical counterpart of

$$\mathbb{E}[|\log(\sigma_\Delta - \log(\sigma_0)|^q]$$

When we plot $\log m(q, \Delta)$ vs $\log(\Delta)$ for different values of q, we found that for each q the historical values lie on a straight line.

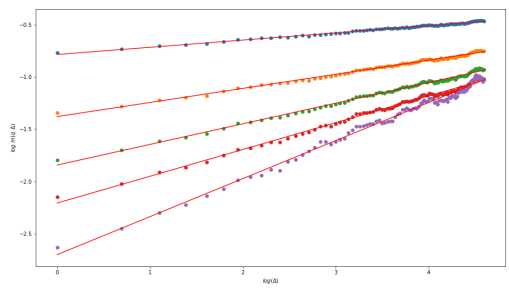

FIGURE 2 – $\log m(q, \Delta)$ as a function of $\log(\Delta)$, S&P

Under stationarity assumptions, this implies that the log-volatility increments respect the following scaling property in expectation :

$$\mathbb{E}[|\log(\sigma_\Delta) - \log(\sigma_0)|^q] = K_q \Delta^{\zeta_q} \tag{3}$$

where $\zeta_q > 0$ is the slope of the line associated to q and plotting ζ_q vs q we notice that ζ_q is a linear function of q : $\zeta_q \sim Hq$

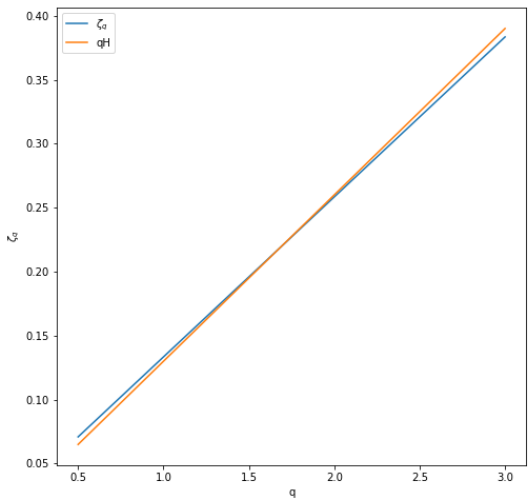

FIGURE 3 – ζ_q and 0.142xq for the S&P

Using a different set of data than the one used in the paper (but on the same index), we obtain the same resultat as the authors : the smoothness parameter H belongs to $[0.08, 2]$.

3.2 Robustness of the estimator

This subject isn't discussed in the paper, this section is inspired by Bennedsen, M., Lunde, A., Pakkanen, M. S. (2016) Decoupling the short-and long-term behavior of stochastic volatility. In this paper they discussed the impact of having to estimate the spot volatility on the estimation of H.

The main idea is that because we are estimating the volatility we are likely to have to deal with data that contains noise. The noise is modeled by a random variable of mean 0, independent of the data and is added to the true value of the volatility. To measure the impact of noise on the estimator used by Gatheral , Jaisson and Rosenbaum we are going to generate paths for a fBM with fixed H and add noise to it. The noise is a normal centred random variable with standard deviation (range of the fbm/250). Then we are going the compare the estimated H for the path with and without noise

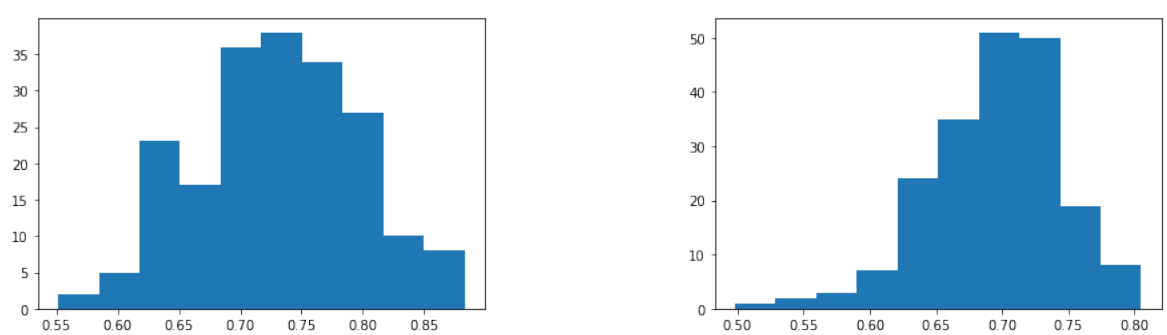

FIGURE 4 – Histogram of estimated H from data without noise (left) and with noise (right) True value H=0.75

We can see from the figures that the estimator doesn't react well to noise. In fact the value of H are changing a lot. The mean value of the data with noise is far from the true value. This is an issue to estimate the exact value of H but the range of the variation is small enough to show that the value of H must be smaller than 1/2

To obtain a better estimation of H, Bennedsen, M., Lunde, A., Pakkanen, M. S. introduced a new estimator for H that is more robust. We are not going to look at the theoretical results about this new estimator we are just going to do the same test that we just did. (see code in Appendix)

$$\widehat{\theta} = (\widehat{a}, \widehat{c}, \widehat{\alpha}) := arg\ inf_{\theta \in \Theta} \sum_{k=1}^{m} (\widehat{\gamma}_2^*(k, \Delta) - a - c(k\Delta)^{2\alpha+1})^2$$

where : $\widehat{\gamma}_2^*(k, \Delta) := \frac{1}{N-k} \sum_{j=1}^{N-k} |\log(\sigma_{(k+j)\Delta}) - \log(\sigma_{(j)\Delta})|^2 (4)$

With this estimator the H we are looking for is equal de $\widehat{\alpha} + 1/2$

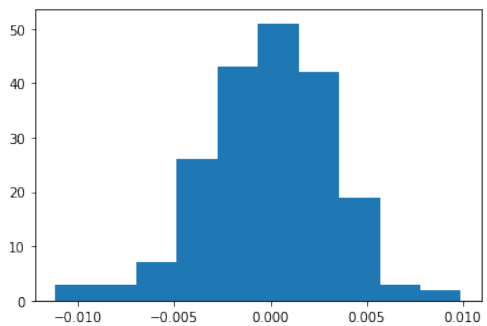

FIGURE 5 – Histogram of the distance between the H estimated with and without noise

We can see that this new estimator reacts better to noise. In average the distance between the estimated H using data with or without noise is 0,0028 and max distance is 0.015. However using this new esimator introduce a new problem : finding the right window parameter, in fact in the formula we can see that we need to specify m. This m has a major impact on the value of the estimated H. This new problem once again prevent us from finding the "True" value of H but as with the previous estimator it is still enough to conccude that H is smaller than 1/2.

3.3 Increments of the log volatility

Once again we did the same analysis as in the paper but with a different data set. We found that they are approximately Gaussian.

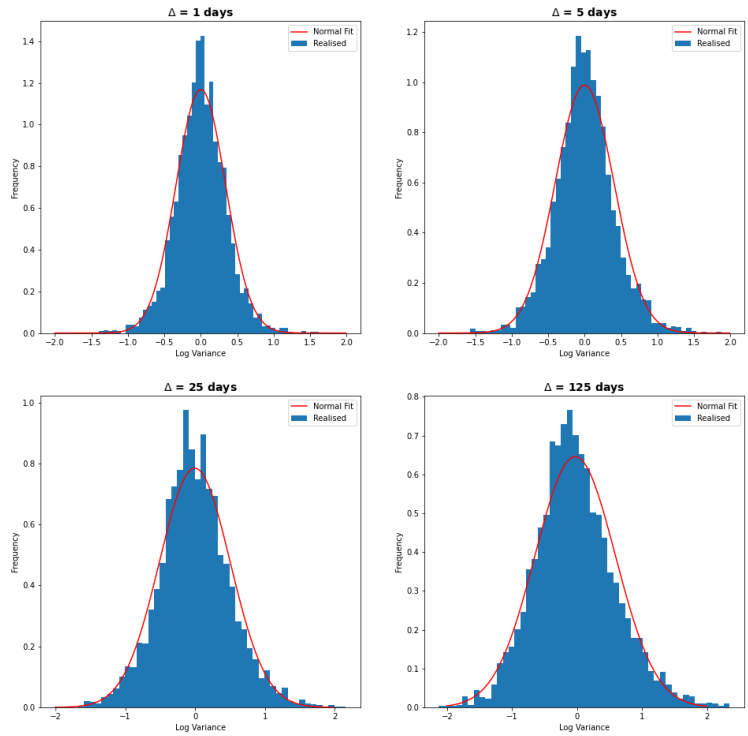

FIGURE 6 – Histograms for various lags Δ of the (overlapping) increments $\log(\sigma_{t+\Delta}) - \log(\sigma_t)$ of the S&P log-volatility ; normal fits in red

To end this part about estimating H note that it is hard to determine if H depends on time or not. In fact as we are dealing with few realisations of random processes the small variation over time could be explained both by the randoness of the realisation or by a dependency to time.

4 A simple model compatible with the empirical smoothness of the volatility

4.1 Specification of the RFSV model

From the above empirically results there are some important facts to retain about the volatility :

- The distribution of increments of log volatility are approximatively normlly distributed.
- The increments of log volatility enjoy a scaling property with constant smoothness parameter

$$\log(\sigma_{t+\Delta}) - \log(\sigma_t) = \nu(W^H_{t+\Delta} - W^H_t) \tag{5}$$

where W^H is a fractional Brownian motion with Hurst parameter equal to the measured smoothness of the volatility and ν is a positive constant. We may of course write (5) under the form

$$\sigma_t = \sigma \exp(\nu W_t) \tag{6}$$

where σ is another positive constant.

But this model is not stationary(mathematical tractability reasonableness) so authors proposed a new model by modelling the log-volatility as a fractional Ornstein-Uhlenbeck process (fOU) with a very long reversion time scale. A stationary fOU process (X_t) is defined as the stationary solution of the stochastic differential equation :

$$dX_t = \nu dW_t^H - \alpha(X_t - m)dt$$

where $m \in \mathbb{R}$, $\nu > 0$ and $\alpha > 0$. The explicit form of the solution is

$$X_t = \nu \int_{-\infty}^{t} e^{-\alpha(t-s} dW_t^H + m \tag{7}$$

They proposed a **Rough Fractional Stochastic Volatility** (RFSV) defined as :

$$\forall t \in [0, T],\ \sigma_t = \exp(Xt). \tag{8}$$

where X_t satisfies (7) for some $m \in \mathbb{R}, \nu > 0, \alpha > 0$ and $H < \frac{1}{2}$.

4.2 RFSV model autocovariance functions

From the definition of the RFSV model, the covariance of $\log(\sigma_t)$(i.e the covariance of X_t) should be linear with respect to Δ^{2H}. To see if this property is satisfied, we plot the S&P autocovariance function of the log-volatility agains't Δ^{2H} and notice that the empirical data is indeed linear with respect to Δ^{2H}. Note in passing that at the time scales we consider, the term $Var[X_t^\alpha]$ is higher than $\frac{1}{2}\nu^2\Delta^{2H}$ in the expression for $Cov[X_t^\alpha, X_{t+\Delta}^\alpha]$.

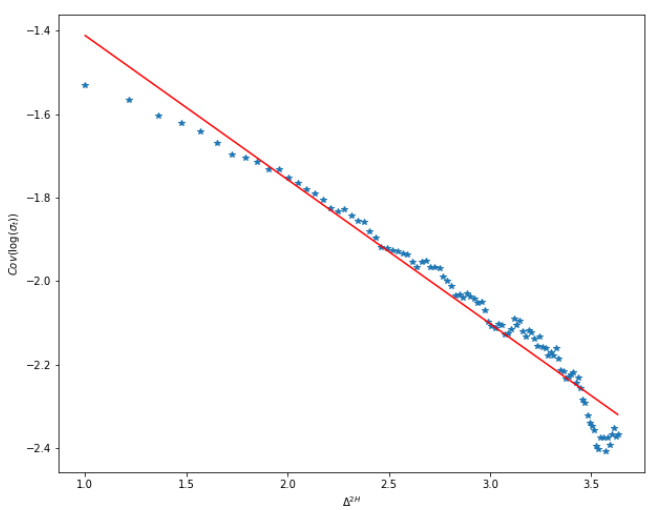

FIGURE 7 – Autocovariance of the log-volatility as a function of Δ^{2H} for H = 0.14, S&P.

An exact formula for the autocovariance function of the log-volatility in the RFSV model was proposed in *Fractional Ornstein- Uhlenbeck processes* (P. Cheridito, H. Kawaguchi, and M. Maejima.) :

$$Cov[\log\sigma_t, \log\sigma_{t+\Delta}] = \frac{H(2H-1)\nu^2}{2\alpha^{2H}} \left\{ e^{-\alpha\Delta}\Gamma(2H-1) + e^{-\alpha\Delta}\int_0^{\alpha\Delta} \frac{e^u}{u^{2-2H}}du + e^{\alpha\Delta}\int_{\alpha\Delta}^{\infty} \frac{e^{-u}}{u^{2-2H}}du \right\} \tag{9}$$

and

$$Var[\log\sigma_t] = \frac{H(2H-1)\nu^2}{2\alpha^{2H}}\Gamma(2H-1) \tag{10}$$

where Γ denotes the Gamma function.

We know that

$$\mathbb{E}[\sigma_t \sigma_{t+\Delta}] = \mathbb{E}[e^{X_t^\alpha} e^{X_{t+\Delta}^\alpha}] = \mathbb{E}[e^{X_t^\alpha + X_{t+\Delta}^\alpha}] = e^{2\mathbb{E}[X_t^\alpha] + 2Var[X_t^\alpha]} e^{-\nu^2 \frac{\Delta^{2H}}{2}} \tag{11}$$

In the RFSV model, from (11) $\log(\mathbb{E}[\sigma_t \sigma_{t+\Delta}])$ should be linear with respect to Δ^{2H}. As shown by the picture bellow, this property is very well satisfied on S&P data :

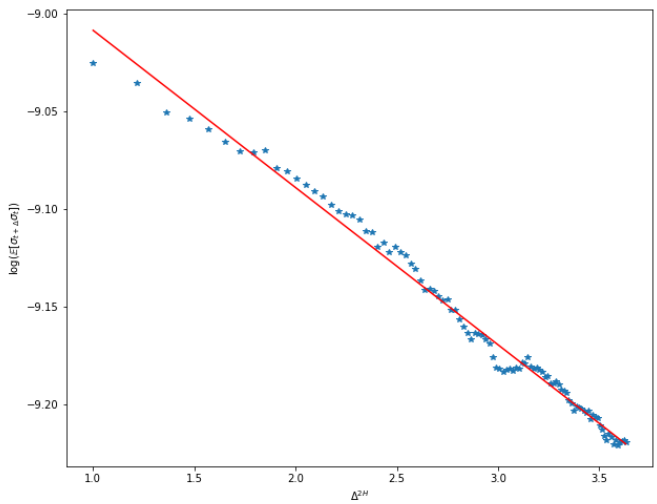

FIGURE 8 – Empirical counterpart of $\log(\mathbb{E}[\sigma_{t+\Delta}\sigma_t])$ as a function of Δ^{2H}, S&P.

4.3 Simulation-based analysis of the RFSV model

The aim of this section is to show that the behaviors concerning smoothness measures observed on empirical data are the same as those obtains on simulated data from the RFSV model. In order to account for the overnight effect, we simulate the volatility σ_t and the efficient price Pt over the whole day. The parameters : $H = 0.14$, $\nu = 0.3$, $m = X_0 = -5$ and $\alpha = 5$ x 10^{-4} are chosen to be consistent with our empirical estimates from previous section. We managed to simulate trajectories for the RFSV unfortunately the computing time is very long (more than 15min for one trajectory). Due to this high computing time we were not able to do our own graph for this part and we are presenting the one from the paper.

First let's just look at one trajectory to compare the shape of the model generetad one with the one from the data.

We see that the shape are very close one to each other.

Now we are going to do the same work as in 3.1 but on simulated data.

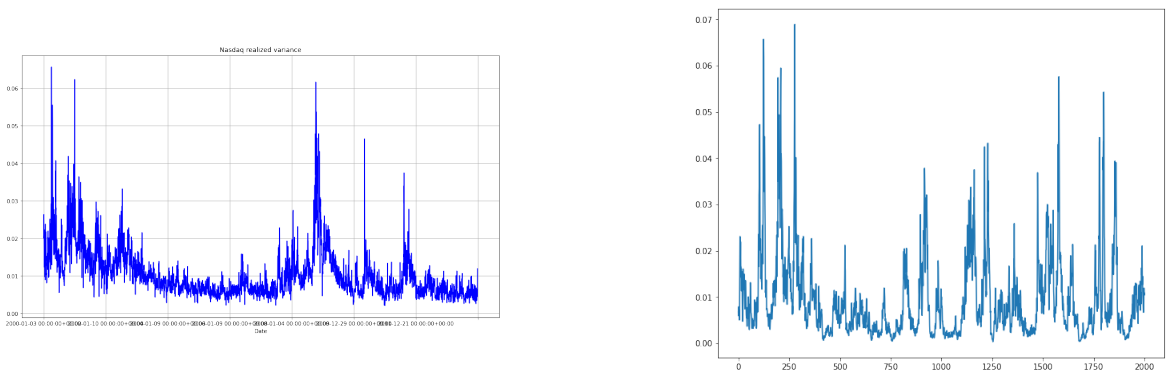

FIGURE 9 – Volatility of the Nasdaq (left) and of the model (right)

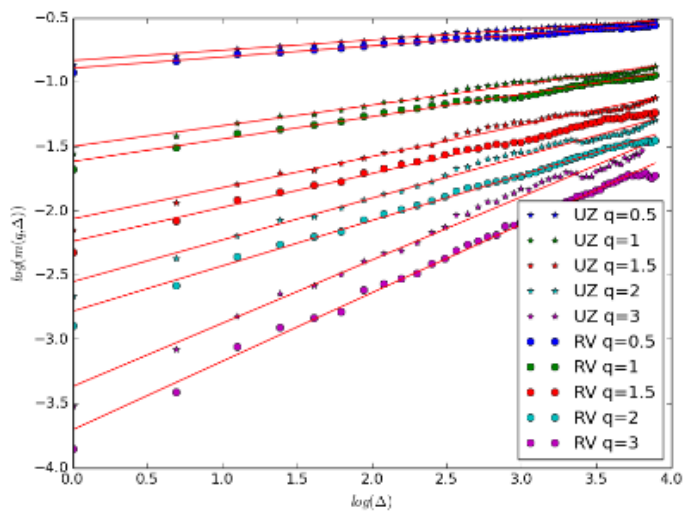

FIGURE 10 – $log(m(q, \Delta))$ as a function of $log(\Delta)$ simulated data (graph from the paper)

On simulated data we obtain the same result as with the real data (figure 2). This is a strong argument for the RSFV validity.

5 Spurious long memory of volatility

One accepted stylized fact is the long memory of volatility, this means that the autocvariance function $Cov[\log(\sigma_t), \log(\sigma_{t+\Delta})]$ go slowly to zero as detla goes to infinity. To be more precise $Cov[\log(\sigma_t), \log(\sigma_{t+\Delta})] \sim \Delta^{-\gamma}$ with $\gamma < 1$ as $\Delta \to \infty$. From (9) and (11) we can see that neither the model nor the data have the right autocovariance function. To undesrtand this difference between the data , the model and the stylezed fact, the authors use the classical method to detect long memory on data simulated with the model. As in previous section due to high computation time we are forced to use the result from the paper as we are not able to do generate a large amount of trajectories.

First they use :

$$V(t) = Var[\int_0^t \sigma_s^2 ds] \tag{12}$$

We can see on the graph that data is close to a staight line with slope 1.86, which according to the method

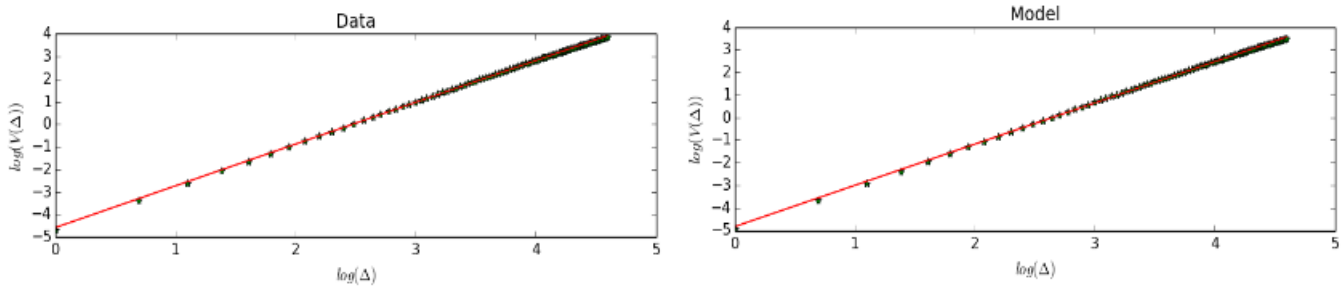

FIGURE 11 – Empirical counterpart of log(V(t)) as a function of log(t) on SP (left) and simulation (right) (graph from the paper)

should imply a power law behvior of the autocorrelation function with power 0.14

A second method is to use wehere L is le lag operator and d= 0.4 :

$$\epsilon_t = (1 - L)^d \log(\sigma_t) \tag{13}$$

Then we compute the autocorrelation function of σ_t

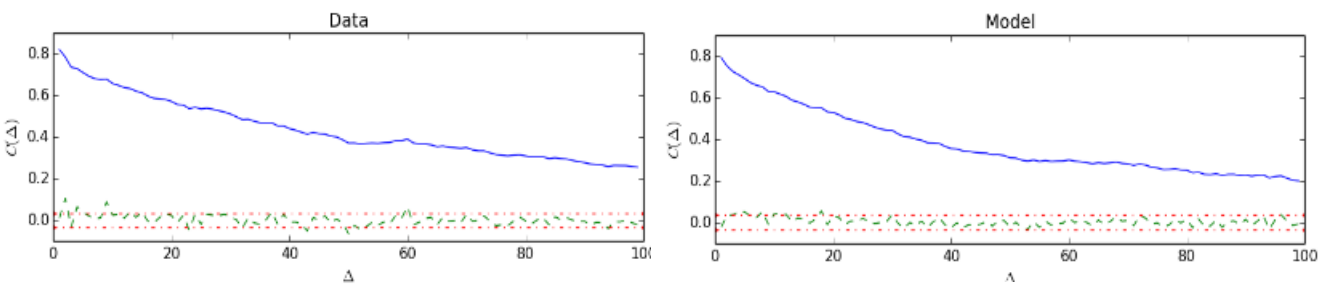

FIGURE 12 – Autocorrelation functions of $\log(\sigma_t)$ (in blue) and ϵ_t (in green) and the Bartlett standard error bands (in red), for SP data (left) and for simulated data (right). (graph from the paper)

Once again the generated data seems to have long term memory according to this method. We can conclude that classical procedures detect long memory for the model and that this stylized fact holds for RFSV.

6 Forecasting using the RFSV model

6.1 Log-volatility prediction

Our prediction formula for log-variance follows :

$$\mathbb{E}[\log \sigma^2_{t+\Delta} | \mathcal{F}_t] = \frac{\cos(H\pi)}{\pi} \Delta^{H+1/2} \int_{-\infty}^{t} \frac{\log(\sigma_s^2)}{(t-s+\Delta)(t-s)^{H+1/2}} ds \tag{14}$$

A comparision between the predictive power of formula (14) and AR/HAR forecasts is proposed. Let recall that for a given integer $p > 0$, the AR(p) and HAR predictors take the following form (where the index i runs over the series of daily volatility estimates) :

- AR(p) :

$$\widehat{\log \sigma^2_{t+\Delta}} = K_0^\Delta + \sum_{i=0}^{p} C_i^\Delta \log(\sigma^2_{t-i}).$$

- HAR :

$$\widehat{\log \sigma^2_{t+\Delta}} = K_0^\Delta + C_0^\Delta \log(\sigma^2_t) + C_5^\Delta \frac{1}{5}\sum_{i=0}^{5} \log(\sigma^2_{t-i}) + C_{20}^\Delta \frac{1}{20}\sum_{i=0}^{20} \log(\sigma^2_{t-i}).$$

$$P = \frac{\sum_{k=500}^{N-\Delta} \left(\log(\sigma^2_{k+\Delta}) - \widehat{\log(\sigma^2_{k+\Delta})}\right)^2}{\sum_{k=500}^{N-\Delta} \left(\log(\sigma^2_{k+\Delta}) - \mathbb{E}[\log \sigma^2_{k+\Delta}]\right)^2}$$

	AR(5)	AR(10)	RFSV
SPX.rv $\Delta = 1$	0.317	0.318	**0.313**
SPX.rv $\Delta = 5$	0.459	0.449	**0.426**
SPX.rv $\Delta = 20$	0.764	0.694	**0.606**

TABLE 1 – Ratio P for the AR and RFSV predictors

We try to replicate the ratio P but we don't get the same results as the ones above.

6.2 Variance prediction

We have the following predictor of the variance in the RFSV model :

$$\widehat{\sigma^2_{t+\Delta}} = \exp\left\{\widehat{\log \sigma^2_{t+\Delta}} + 2c\nu^2\Delta^{2H}\right\} \tag{15}$$

where $\widehat{\log \sigma^2_{t+\Delta}}$ is the estimator from Section 6.1, $c = \frac{\Gamma(3/2-H)}{\Gamma(H+1/2)\Gamma(2-2H)}$ and ν^2 is equal to the exponential of the intercept in the linear regression of $\log(m(2,\Delta))$ against $\log(\Delta)$.

7 The microstructural foundations

We want to understand the irregularity of the volatility by taking a look at the order flow through a Hawkes Process. Let consider a time period $[0,t]$ and denote N_t the number of transaction that occurs between 0 and t(a Hawkes process). Then the intensity λ_t at time t of this process takes the form :

$$\lambda_t = \mu + \sum_{0<J_i<t} \phi(t - J_i),$$

where J_i are the past jump times, μ a postive constant and ϕ is a non negative deterministic function called kernel. In concrete terms, each order implies $\int_0^\infty \phi(s)ds = ||\phi||_1$ other orders. This happen because market participants react to other participants orders because of electronic high frequency trading : This is called **Market endogeneity**.
In a infinite horizon time, we will have $||\phi||_1 + ||\phi||_1^2 + ||\phi||_1^3 + ... + = \sum_{i=1}^\infty ||\phi||_1^i$. Since we don't want this sum to explose, we need to have $||\phi||_1 < 1$ which lead our total number of participants to be equal to $1 + \frac{||\phi||_1}{1-||\phi||_1}$.
So the average proportion of endogenously triggered events is equal to $E = \frac{\frac{||\phi||_1}{1-||\phi||_1}}{1+\frac{||\phi||_1}{1-||\phi||_1}} = ||\phi||_1$.
When authors tried to calibrate a Hawkes process on high frequency data, they note that the L^1 norm of ϕ ($||\phi||_1$) is closed to one, witch mean that **the endogeneity of the market is very strong** : there is a lot of orders that occurs because of existing orders.

8 Conclusion

To conclude, we can say that : the main result we found in this article is the fact that the distributions of increments of log-volatility are approximately Gaussian, precisely can be modeled as fBm :

$$log(\sigma_{t+\Delta}) - log(\sigma_t) = \nu(W^H_{t+\Delta} - W^H_t, \quad H \in (O.06, 0.2)$$

One can notice also the monofractal scaling relationship :

$$\mathbb{E}[|log(\sigma_\Delta) - log(\sigma_0)|^q] = K_q \nu^q \Delta^{qH}$$

where H measure the smoothness characteristic of the underlying volatility process.
Moreover, the autors can deduce the long memory stylized fact of volatility despite taking H less than 1/2. It is shown by applying standard statistical estimators to volatility time series.
To apply the RFSV model, using data different of what authors have used, we tried to forecast volatility and variance at various time scale.
But how can we explain such irregalarity of volatility ?
By studying, the scaling limit of Hawkes process, rough volatility could result of the **high degree of endogenity of the market** and **the order splitting**.
Finally, we can extend our work by studying the volatility behaviour when H is **negative** or by applying our result to option pricing, using for instance the **Rough Heston model**.

9 Appendix

9.1 Theorem : Kolmogorov continuity theorem

Assume that for a stochastic process : $(X_t)_{t\geq 0}, \exists K > 0, p > 0, \beta > 0 | \forall t \geq 0, s \geq 0$:

$$\mathbb{E}[|X_t - X_s|^p] \leq K|t-s|^{1+\beta}$$

Then the process X has a continuous modification, i.e a process $(\tilde{X}_t)_{t\geq 0} | \tilde{X} \in C[0, +\infty[, \forall t \geq 0, P(X_t = \tilde{X}_t) = 1$. Moreover, $\forall \gamma \in (0, \frac{\beta}{p}, T > 0$, the process $\tilde{X} is \gamma$-Holder continous on $[0, T], i.e$:

$$\sup_{0\leq s\leq t\leq T} \frac{|X_t - X_s|}{(t-s)^\gamma} < +\infty$$

9.2 Corollary

The fBm has continous modification : for any $\gamma \in (0, H)$, this modification is γ-Holder continous.
Proof : Since $W^H_t - W^H_s$ is centerd Gaussian with variance $|t-s|^H$, we have $\mathbb{E}[|W^H_t - W^H_s|^p] = K_p|t-s|^{pH}$. Therefore, taking $p > \frac{1}{H}$, we get the existence of continous modification. Also : the Holder continuity, with exponent $\gamma \in (0, H - \frac{1}{p})$. So we just have to choose p sufficiently large.